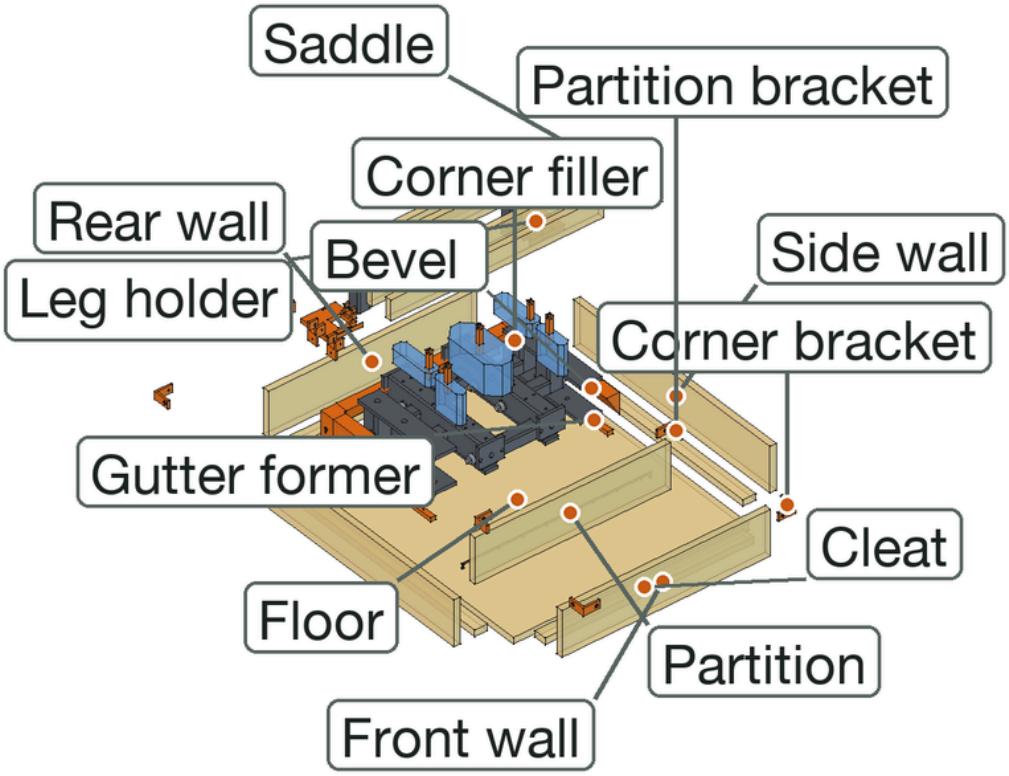
Base mold

DIY CNC epoxy-granite mill · October 2026

How to make the base mold. Drilling templates: `templates/base-templates.pdf`. The sheet layout for the boards is `docs/src/boards/sheet-1.svg`; `docs/src/make_pdfs.py` also makes `molds-complete.pdf`, all the molds in one (not kept in the repo).

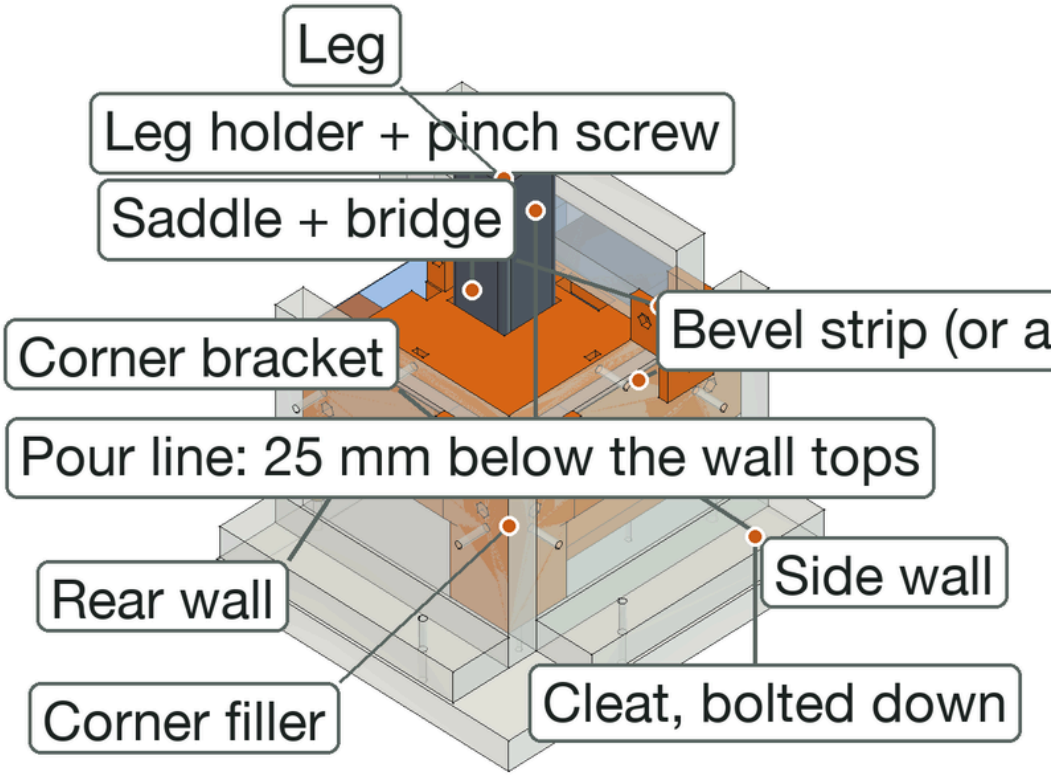


In short: melamine boards bolt together with printed brackets. Every printed piece bolts or pins to its board from outside. The walls stand 25 mm above the pour line, so no bolt goes through a face the mix touches. Nothing is glued, and it all comes apart for the next casting.

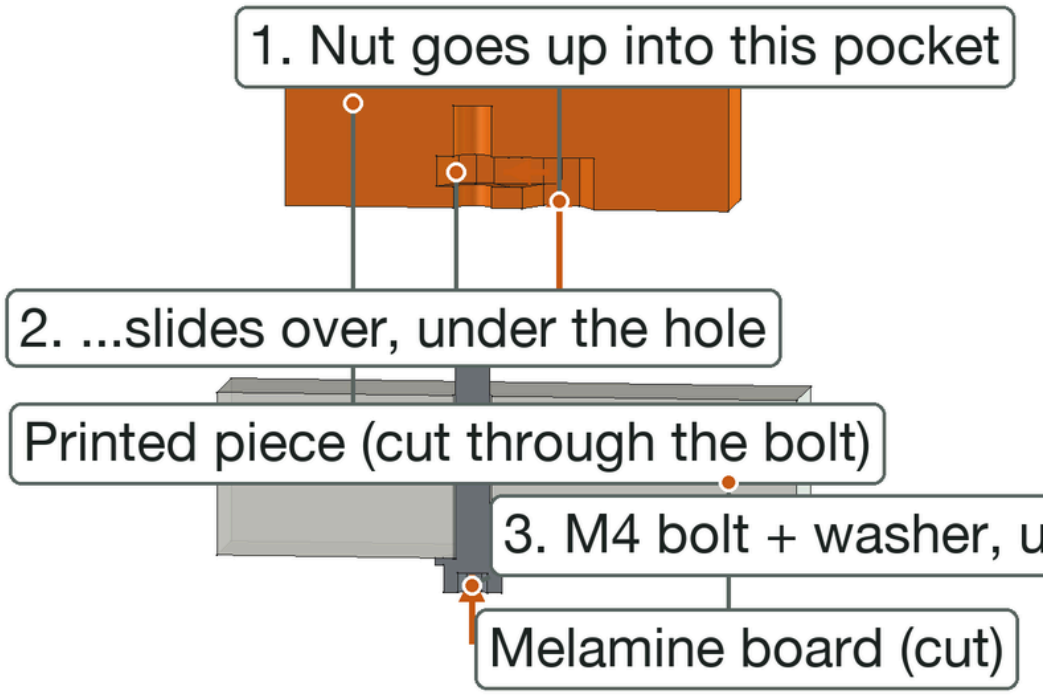
Order of work: print → cut the boards → mark them with the templates and drill → cut the foam → wax → assemble → pour → take apart.

1. How it works

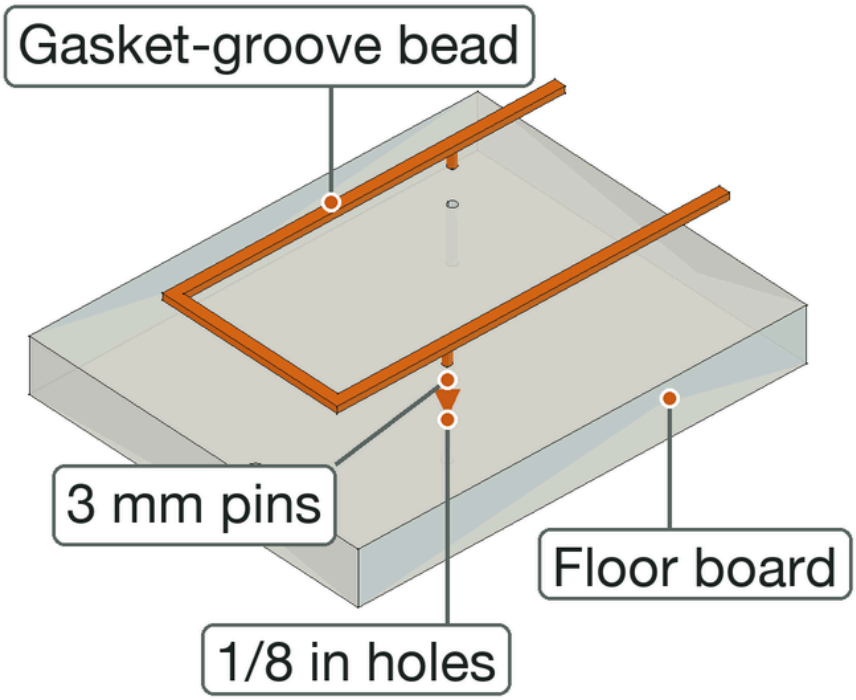
- **Walls** stand on the floor against bolted-down **cleats**; **corner brackets** square the tops. All the bolts go through the top 25 mm, above the pour line.



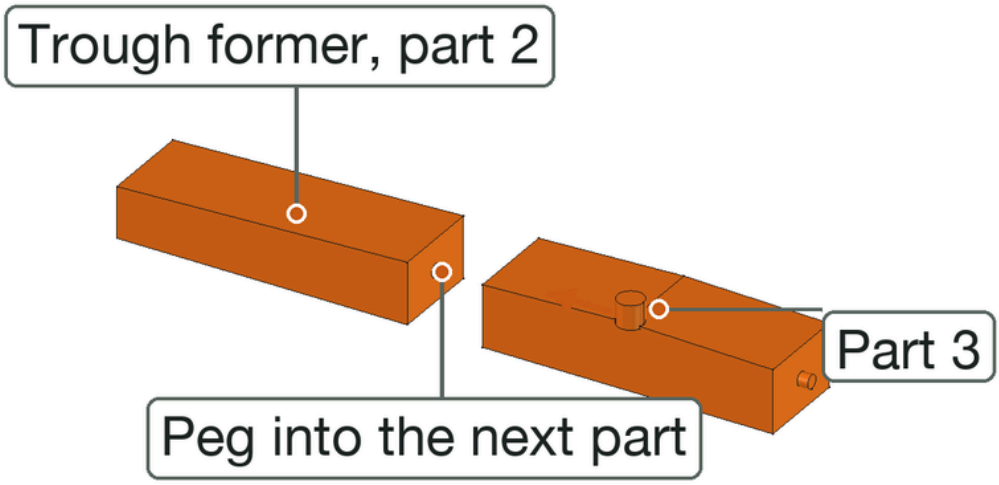
- **Printed pieces** have a nut built in: it goes into a pocket on the board side and slides under the hole; an M4 bolt comes through the board from outside.



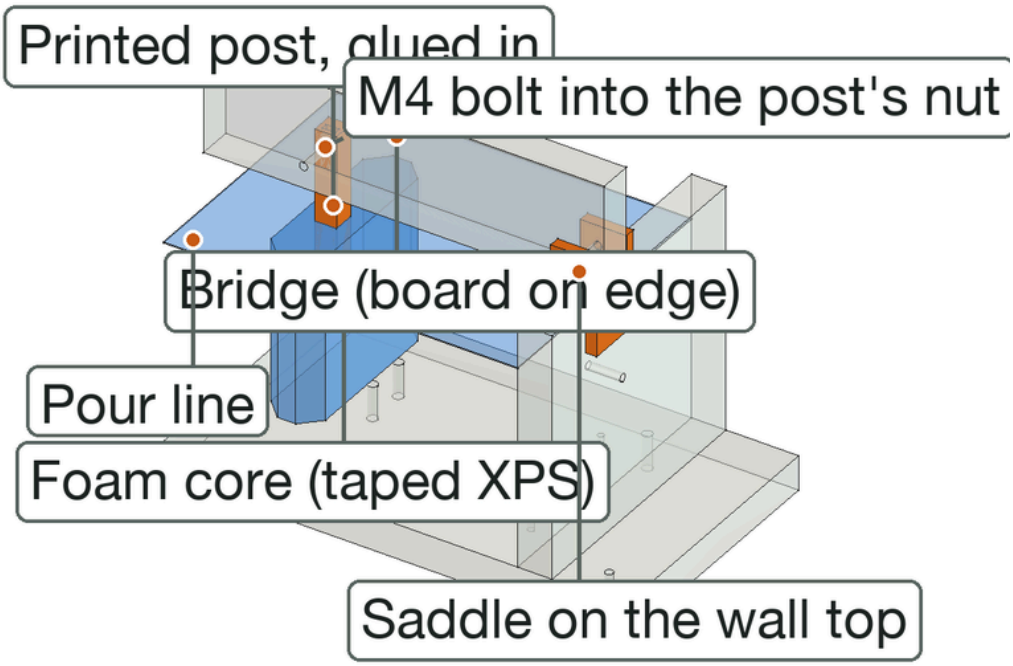
- **Thin pieces** (gasket-groove beads) have 3 mm pins that push into 1/8 in holes.



- **Long pieces** are printed in parts that peg together; each part bolts down on its own.



- **Cores** are taped foam blocks with a printed post that bolts to a bridge across the mold.



- **Fits** are 0.15 mm a side, and drawn for 19.05 mm board (Home Depot's 3/4 in melamine is a true 0.750 in). Measure yours; if it differs, set `mold_board` in `model/build_cnc.py` and re-run it.

2. What you need

Material	Amount	Notes
Melamine board, 3/4 in	about 1.3 m2	all the molds come out of 1 sheet, 4 x 8 ft
1x2 lumber (furring strip is fine)	2 x 8 ft 1x2 (2.7 m)	the cleats: they lie on the floor outside the walls
XPS foam board, 2 in	1 sheet, 2 x 8 ft	the cores
PETG	1.39 kg	sliced; everything printed for the machine fits in two 1 kg rolls
Packing tape, double-sided carpet tape	1 roll each	wrapping and holding the foam
Paste wax and PVA	1 each	release
3/8 in quarter-round PVC molding	a few meters	the rounded outside edges

Hardware	Qty
M3 nut	20
M3 washer	20
M3 x 25 bolt	20
M4 nut	116
M4 washer	112
M4 x 20 bolt (leg pinch screw)	4
M4 x 30 bolt	68

Hardware	Qty
M4 x 35 bolt	28
M4 x 45 bolt	16
M5 x 20 screw, greased (holds an insert)	17
M5 x 30 screw, greased (holds a rail bar or the BK pad)	12
M6 x 30 screw, greased (holds a support pad)	8

Add about 10 % spare small bolts, nuts and washers. **Tools:** drill and brad-point bits (sizes in section 5), a 2-1/2 in hole saw, a square, clamps, a 7 mm nut driver, a 3 mm hex key, an awl, a fine saw or hot wire for the foam.

3. Print

74 prints, 1.39 kg, about 89 hours (sliced; settings in the print guide). Print one corner bracket and one saddle first and check them on a board offcut. Press or slide the nuts in before the pieces go on the boards.

File	Qty	g	Time each
`base-box/kit-corner-bracket.stl`	4	33	39m 23s
`base-box/kit-partition-bracket.stl`	2	14	35m 28s
`base-box/kit-saddle.stl`	6	58	46m 3s
`base-box/kit-leg-holder.stl`	4	166	2h 30m 55s
`base-rear/former-gutter-L-part1of3.stl`	1	8	35m 6s
`base-rear/former-gutter-L-part2of3.stl`	1	8	33m 40s
`base-rear/former-gutter-L-part3of3.stl`	1	7	31m 15s
`base-rear/former-gutter-R-part1of3.stl`	1	8	35m 6s
`base-rear/former-gutter-R-part2of3.stl`	1	8	33m 40s
`base-rear/former-gutter-R-part3of3.stl`	1	7	31m 15s
`base-rear/filler-corner-L.stl`	1	29	2h 10m 27s
`base-rear/filler-bevel-L-part1of2.stl`	1	50	2h 39m 10s
`base-rear/filler-bevel-L-part2of2.stl`	1	50	2h 39m 41s
`base-rear/filler-corner-R.stl`	1	29	2h 9m 35s
`base-rear/filler-bevel-R-part1of2.stl`	1	50	2h 34m 47s
`base-rear/filler-bevel-R-part2of2.stl`	1	50	2h 35m 51s
`base-rear/filler-bevel-end-part1of3.stl`	1	50	2h 36m 44s
`base-rear/filler-bevel-end-part2of3.stl`	1	50	2h 37m 9s
`base-rear/filler-bevel-end-part3of3.stl`	1	50	2h 36m 53s
`base-front/former-gutter-L-part1of2.stl`	1	11	49m 24s
`base-front/former-gutter-L-part2of2.stl`	1	10	43m 44s
`base-front/former-gutter-R-part1of2.stl`	1	11	49m 24s

File	Qty	g	Time each
base-front/former-gutter-R-part2of2.stl	1	10	43m 43s
base-front/former-trough-part1of5.stl	1	19	1h 12m 26s
base-front/former-trough-part2of5.stl	1	22	1h 21m 44s
base-front/former-trough-part3of5.stl	1	24	1h 31m 38s
base-front/former-trough-part4of5.stl	1	22	1h 21m 55s
base-front/former-trough-part5of5.stl	1	20	1h 11m 51s
base-front/filler-corner-L.stl	1	29	2h 10m 58s
base-front/filler-bevel-L-part1of2.stl	1	46	2h 25m 10s
base-front/filler-bevel-L-part2of2.stl	1	46	2h 25m 21s
base-front/filler-corner-R.stl	1	29	2h 8m 15s
base-front/filler-bevel-R-part1of2.stl	1	46	2h 21m 6s
base-front/filler-bevel-R-part2of2.stl	1	46	2h 21m 45s
base-front/filler-bevel-end-part1of3.stl	1	50	2h 32m 21s
base-front/filler-bevel-end-part2of3.stl	1	50	2h 33m 43s
base-front/filler-bevel-end-part3of3.stl	1	50	2h 33m 17s
base-rear/former-split-gasket-groove-part1of4.stl	1	2	9m 41s
base-rear/former-split-gasket-groove-part2of4.stl	1	1	8m 49s
base-rear/former-split-gasket-groove-part3of4.stl	1	1	8m 49s
base-rear/former-split-gasket-groove-part4of4.stl	1	2	9m 42s
base-front/core-center-a-post.stl	1	5	44m 56s
base-front/core-center-b-post.stl	1	5	44m 56s
base-rear/core-center-post.stl	1	5	44m 56s
base-front/core-outboard-L-post.stl	1	5	44m 53s
base-rear/core-foot-b-L-post.stl	1	5	44m 53s
base-rear/core-knee-L-post.stl	1	5	44m 53s
base-front/core-outboard-R-post.stl	1	5	44m 53s
base-rear/core-foot-b-R-post.stl	1	5	44m 53s
base-rear/core-knee-R-post.stl	1	5	44m 53s
base-front/plug-draw-pocket-L.stl	1	12	46m 16s
base-front/plug-draw-pocket-R.stl	1	12	46m 16s
base-rear/plug-draw-pocket-L.stl	1	5	21m 23s
base-rear/plug-draw-pocket-R.stl	1	5	21m 23s
tools/kit-pour-gauge.stl	2	9	24m 33s
tools/kit-screed-shoe.stl	2	20	44m 42s
base-front/cast-in-leg-front-l-plug.stl	2	8	24m 14s

File	Qty	g	Time each
`base-rear/cast-in-leg-rear-l-plug.stl`	2	8	24m 14s

4. Cut the boards

Have the store rip the strips; crosscut at home. Write each board's name on its outside face, and mark the cast face (the melamine face the mix touches). The cleats are cut from 1x2 lumber, not the sheet; the bridges and struts come out of the sheet's offcuts.

Board	Qty	Size (mm)	Notes
Base floor	1	700 x 843	both halves are cast on it
Base front wall	1	620 x 135	
Base rear wall	1	620 x 135	
Base side wall, left	1	724 x 135	
Base side wall, right	1	724 x 135	
Base partition	1	582 x 135	rear half only; comes out before the front half
Base partition strut	2	326 x 60	offcut, stands on edge; holds the partition foot (rear half only)
Base cleat, front	1	620 x 38	1x2 lumber; flat on the floor
Base cleat, rear	1	620 x 38	1x2 lumber; flat on the floor
Base cleat, left	1	724 x 38	1x2 lumber; flat on the floor
Base cleat, right	1	724 x 38	1x2 lumber; flat on the floor
Base bridge y=-230	1	633 x 60	
Base bridge y=-130	1	633 x 60	
Base bridge y=-60	1	633 x 60	
Base bridge y=100	1	633 x 60	
Base bridge y=170	1	633 x 60	
Base bridge y=280	1	633 x 60	
Base leg bridge y=-280	1	402 x 60	
Base leg bridge y=318	1	402 x 60	

5. Mark and drill

1. Print the drilling templates (`templates/base-templates.pdf`) at **actual size** and check the 100 mm square.
2. Tape each board's pages onto it: printed edges on the board's edges, grid lines lined up.
3. Awl through every crosshair. Pencil the dashed **pour line** on the walls.
4. Drill from the marked face, at the size the color legend gives. Pilot the big ones; the 63.5 mm holes are a 2-1/2 in hole saw.

The appendix has every board as a drawing and a table, if you would rather measure.

Hole	Drill	For
3.2 mm	1/8 in	split-gasket-groove
3.4 mm	9/64 in	corner-L, corner-R, gutter-L, gutter-R
4.5 mm	11/64 in	bevel-L, bevel-R, bevel-end, cleat, core post, corner bracket, corner-L, corner-R, draw-pocket-L, draw-pocket-R, gutter-L, gutter-R, leg holder, saddle, trough
5.5 mm	7/32 in	M5 insert screw, M5 screw into the steel
6.5 mm	17/64 in	M6 screw into a pad
9.6 mm	25/64 in	3/8 dowel, 3/8 dowel pin

6. Cut the foam

Fine saw, hot wire or a long knife. Stack and glue two layers of 2 in foam for the 90 mm cores. Wrap each piece smooth in packing tape. Cut a snug 12 mm square hole 30 mm deep for each core's post and glue the post in.

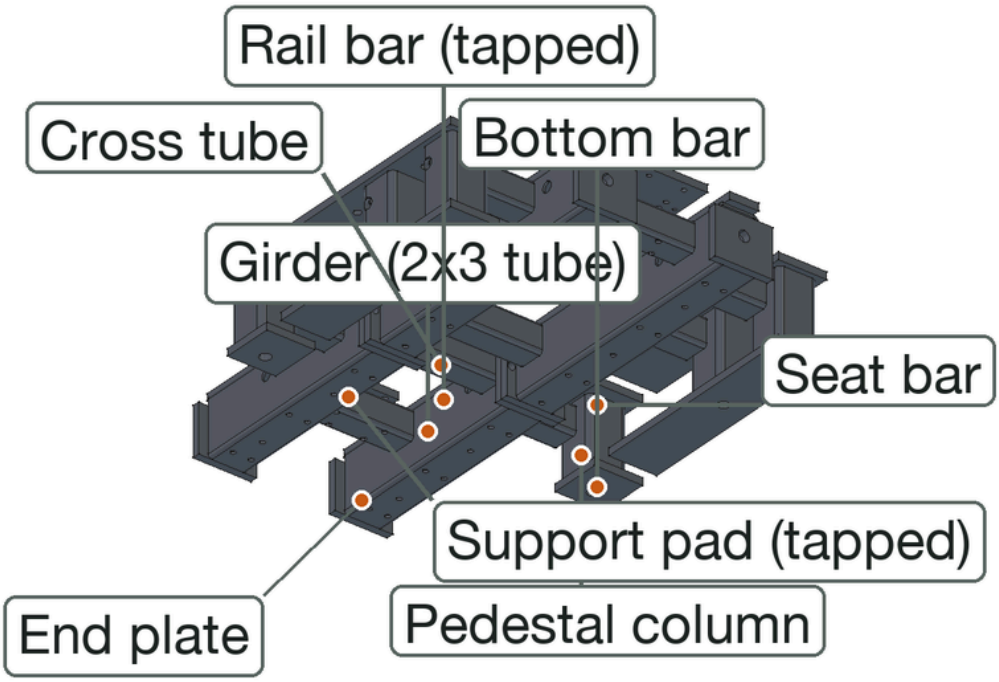
Casting	Piece	Qty	Size and how
base-front	center-a	1	XPS block 100 x 73 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	center-b	1	XPS block 100 x 72 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	center	1	XPS block 88 x 190 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	outboard-L	1	XPS block 93 x 148 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	foot-b-L	1	XPS block 30 x 99 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	knee-L	1	XPS block 41 x 140 x 40 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	outboard-R	1	XPS block 93 x 148 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	foot-b-R	1	XPS block 30 x 99 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	knee-R	1	XPS block 41 x 140 x 40 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
## 7. Wax			

- 3-5 coats of paste wax on every face the mix touches, boards and removable printed pieces, then PVA. Don't wax pieces that stay in (core posts, cast-in parts): scuff and degrease them.
- Glue quarter-round into the floor-to-wall corners that make outside edges; silicone fillets elsewhere.
- Grease every screw, bolt and rod that comes out of the casting.

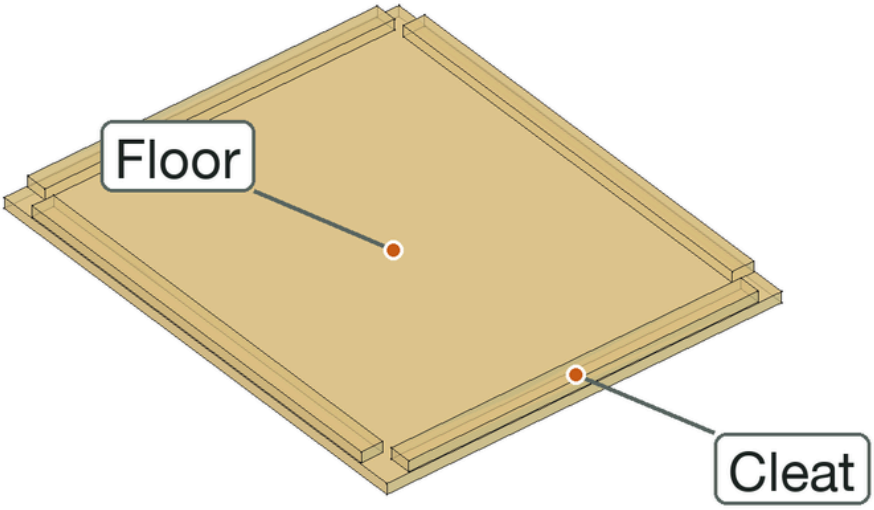
8. Assemble

Base, rear half (first)

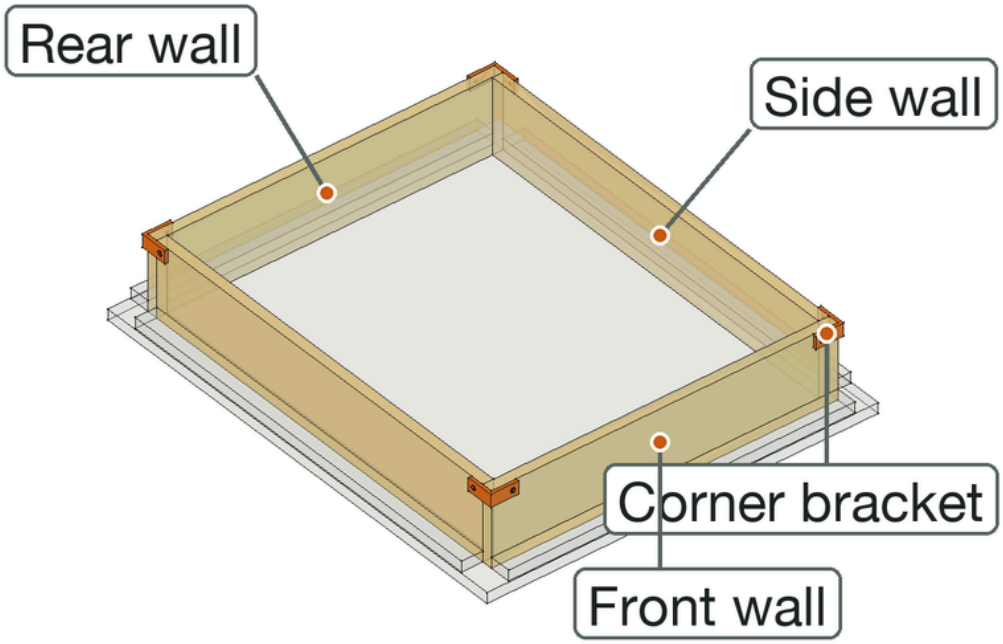
Cast upside down, top face on the floor; shown the right way up. New parts in color. Bolt each half's steel grid together first (the build guide, section 1), then it goes in as one piece.



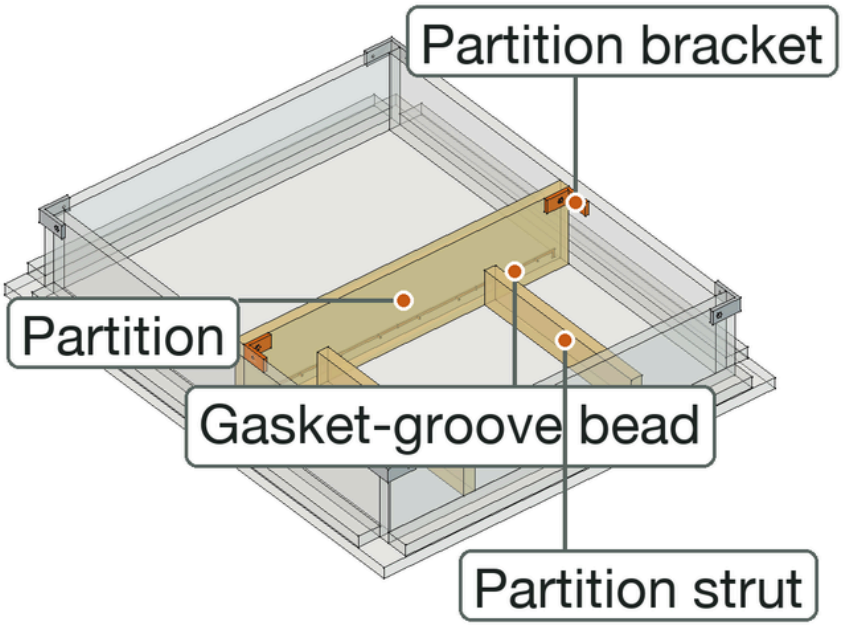
1. Floor on three battens. Bolt the cleats down.



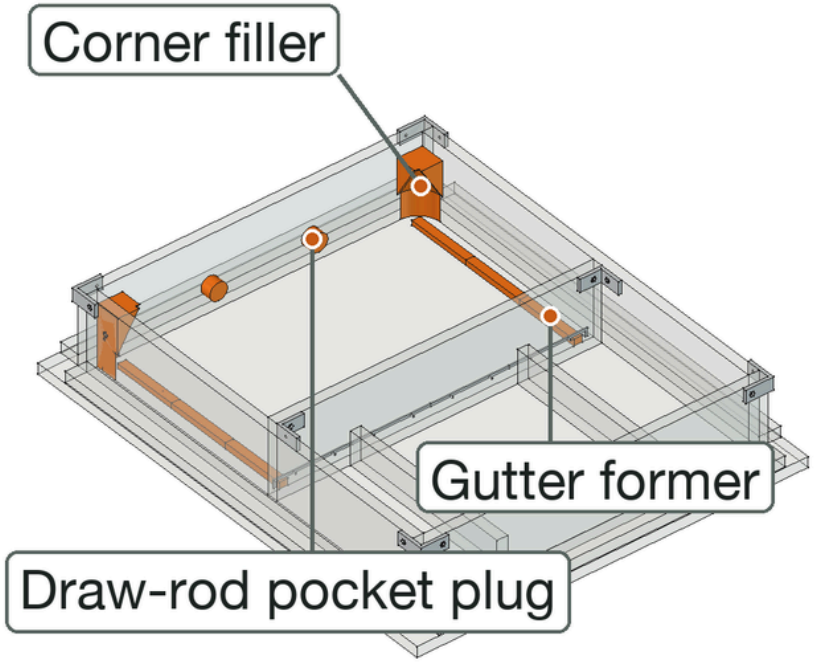
2. Walls against the cleats; corner brackets at the top. Diagonals equal within 2 mm, then tighten.



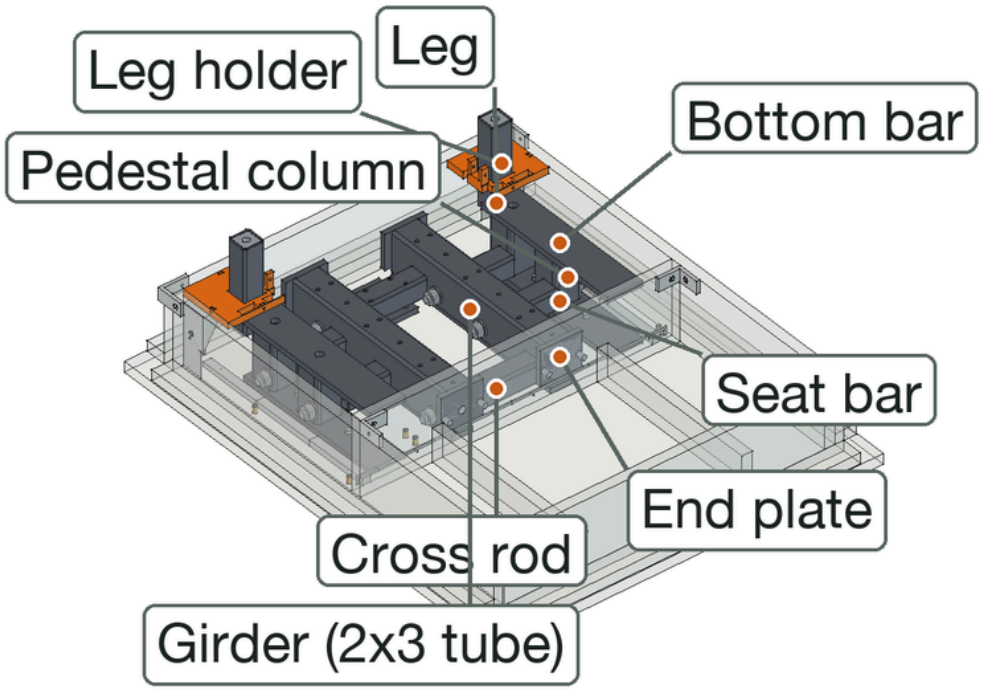
3. Partition between the side walls: two brackets, two struts to the front wall. Beads on its pins.



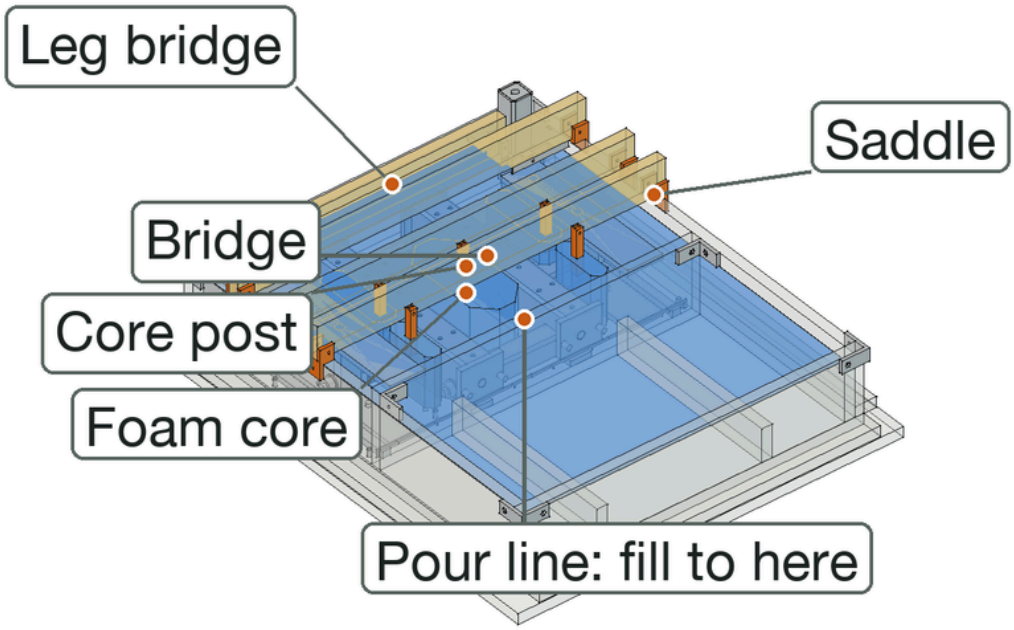
4. Gutter formers, corner fillers and the draw-rod pocket plugs on. Bevel strips on the side and rear walls, their 1-1/2 in face flush with the pour line.



5. The steel grid in, bolted up and upside down: its rail bars and the BK pad screwed to the floor (greased M5 and M6), its seats over the dowels, the split end plates against the partition with their dowel pins through it. Tape over the holes in the pedestals' bottom bars. Inserts; leg holders, legs set to height, pinch screws tight.

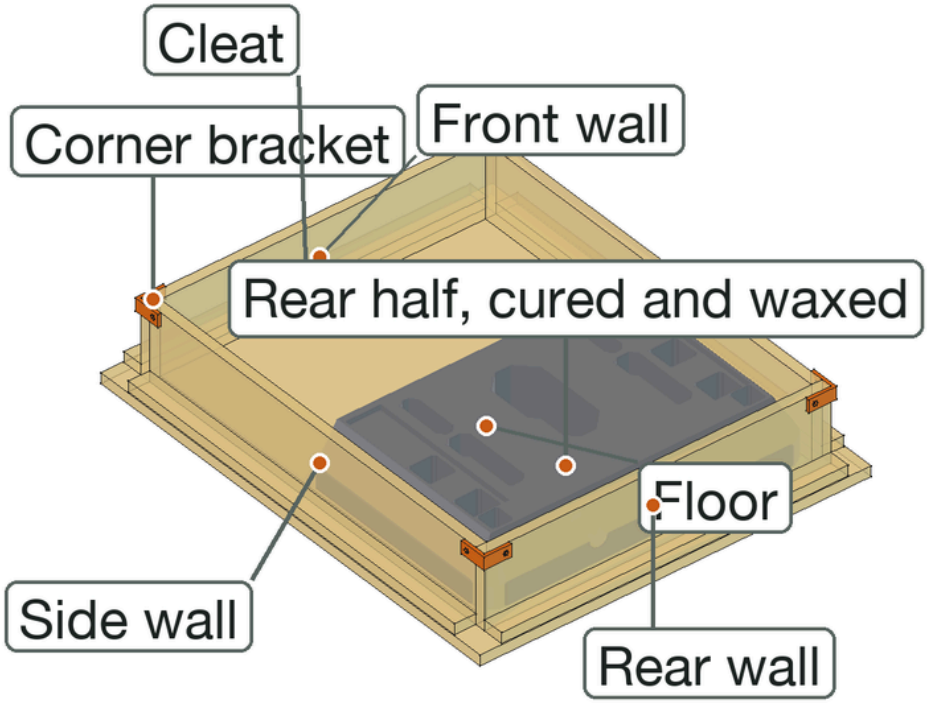


6. Pour the 20 mm skin. Foam cores on it between the steel, bridges into the saddles, each post bolted to its bridge. Fill to the pour line, packing the mix up under the bevel strips.

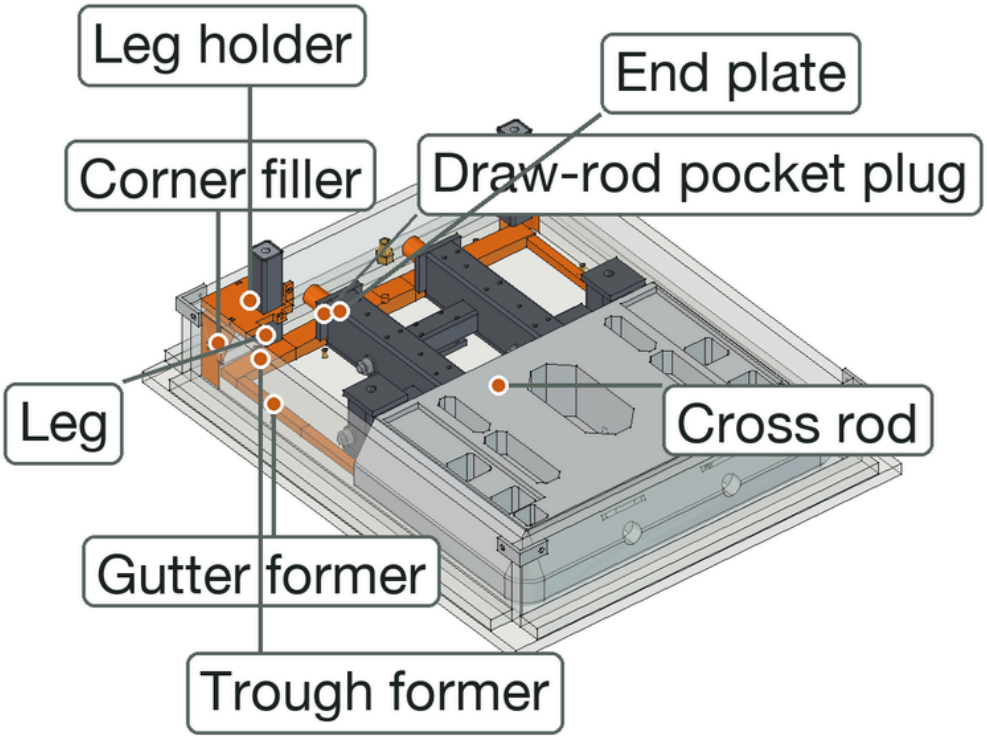


Base, front half (against the rear half)

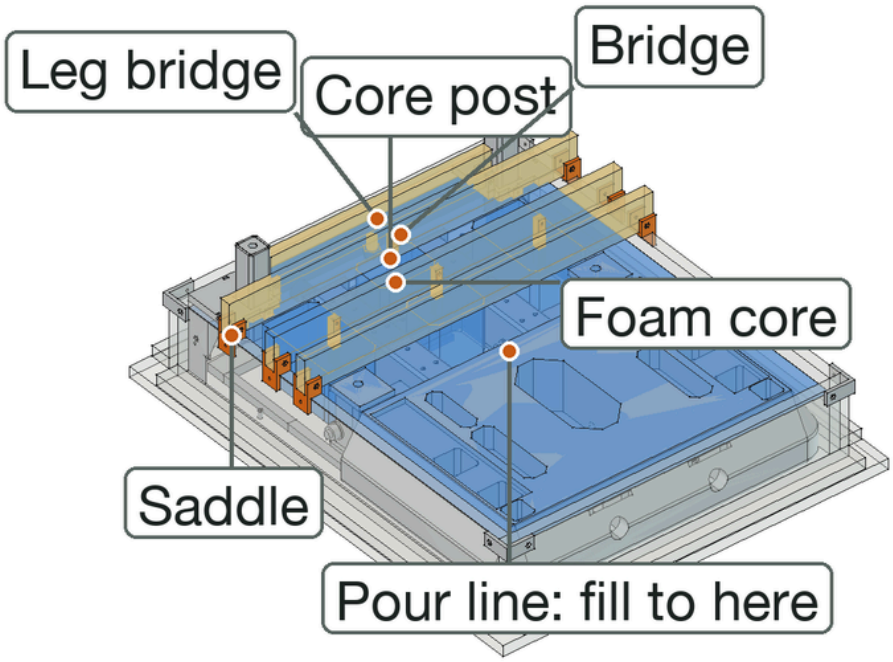
1. After the cure: bridges, partition and struts off. Pull the beads, tape the groove, and wax the rear half's split face, end plates and dowel pins.



2. Trough and gutter formers, corner fillers, bevel strips and pocket plugs on; the front grid in, its rail bars and BF pad screwed down and its split end plates onto the dowel pins; inserts and legs.

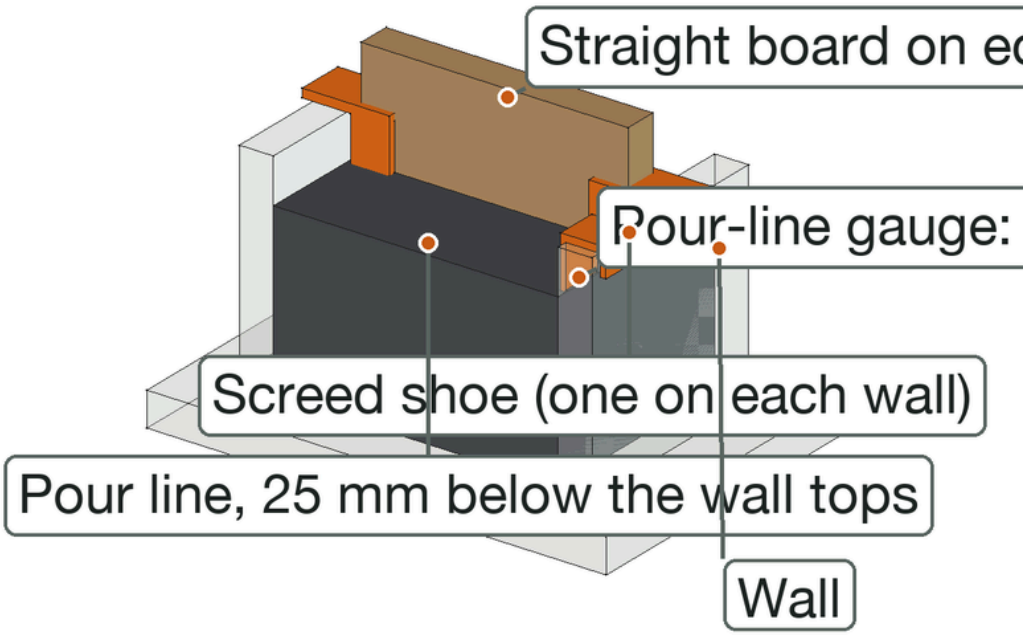


3. Skin, cores, bridges, fill (packed under the bevel strips). After the cure, take it apart and split the halves with wedges.



9. Pour

- Fill to the **pour line**, 25 mm below the wall tops (pencilled from the templates, or with the printed pour-line gauge clipped over a wall).
- Vibrate each layer until it stops bubbling; check the brackets halfway through.
- Strike the top off with a straight board in the two printed **screed shoes**: they hang it from the wall tops with its bottom edge on the pour line.



10. Take apart

1. After the full cure: bridges and saddles off, then corner brackets, then cleats; pull the walls off.
2. Screws, bolts and rods out from outside; lift the casting off the floor.
3. Unbolt and pull the formers and fillers.
4. Scrape and re-wax the boards; keep each mold's kit in a box. The kit is reused.

11. If something doesn't fit

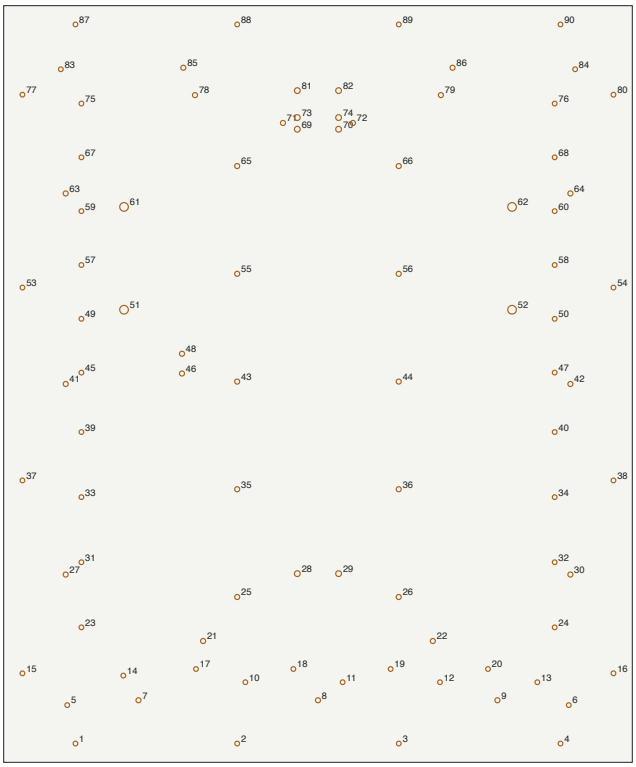
Problem	Fix
Nut won't slide under the hole	Clean the pocket with a nut on a bolt; warm the nut
Bracket or saddle loose on the board	Board thinner than set: tape shim, or set `mold_board` and reprint
Pin won't go in	Run the 1/8 in bit through again; trim the pin tip
Foam core moves	Tighten its post bolt; carpet tape under it

Problem	Fix
Printed part won't go on or into a steel tube	File the weld seam; sand the part. Several tight: caliper the tube, put its sizes in `frame_tube` / `post_tube`, re-run and reprint
Printed part loose on a tube	A wrap of tape round the part
Leak at a corner	Silicone fillet inside before waxing; check the brackets and diagonals
Former stuck after curing	Bolts out first, then a dowel and a light tap

Appendix: hole maps

Every board as seen from the face you mark: x from the left edge, y from the bottom edge (for a wall, the edge on the floor). The templates carry the same holes at full size.

Base floor



0,0 · 700 x 843 mm · seen from the cast side

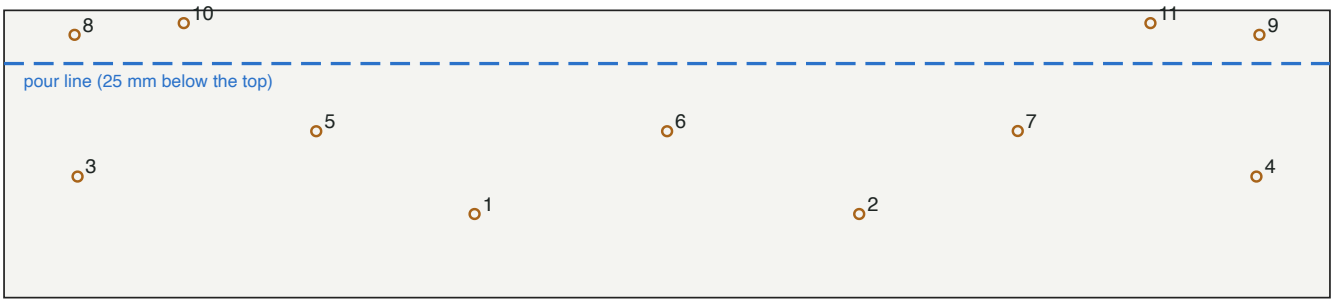
#	x	y	Hole	For
1	80	21	4.5	cleat
2	260	21	4.5	cleat
3	440	21	4.5	cleat
4	620	21	4.5	cleat
5	71	64	3.4	corner-R
6	630	64	3.4	corner-L
7	150	69	5.5	M5 insert screw

#	x	y	Hole	For
8	350	69	5.5	M5 insert screw
9	550	69	5.5	M5 insert screw
10	269	89	4.5	trough
11	378	89	4.5	trough
12	486	89	4.5	trough
13	595	89	4.5	trough
14	133	97	4.5	trough
15	21	99	4.5	cleat
16	680	99	4.5	cleat
17	214	104	4.5	trough
18	323	104	4.5	trough
19	431	104	4.5	trough
20	540	104	4.5	trough
21	222	135	5.5	M5 insert screw
22	478	135	5.5	M5 insert screw
23	87	150	4.5	gutter-R
24	614	150	4.5	gutter-L
25	260	184	5.5	M5 screw into the steel
26	440	184	5.5	M5 screw into the steel
27	69	209	5.5	M5 insert screw
28	327	210	6.5	M6 screw into a pad
29	373	210	6.5	M6 screw into a pad
30	631	209	5.5	M5 insert screw
31	87	223	4.5	gutter-R
32	614	223	4.5	gutter-L
33	87	295	3.4	gutter-R
34	614	295	3.4	gutter-L
35	260	304	5.5	M5 screw into the steel
36	440	304	5.5	M5 screw into the steel
37	21	314	4.5	cleat
38	680	314	4.5	cleat
39	87	368	3.4	gutter-R
40	614	368	3.4	gutter-L
41	69	421	5.5	M5 insert screw
42	631	421	5.5	M5 insert screw

#	x	y	Hole	For
43	260	424	5.5	M5 screw into the steel
44	440	424	5.5	M5 screw into the steel
45	87	434	3.4	gutter-R
46	199	433	5.5	M5 insert screw
47	614	434	3.4	gutter-L
48	199	455	5.5	M5 insert screw
49	87	494	3.4	gutter-R
50	614	494	3.4	gutter-L
51	134	504	9.6	3/8 dowel
52	566	504	9.6	3/8 dowel
53	21	529	4.5	cleat
54	680	529	4.5	cleat
55	260	544	5.5	M5 screw into the steel
56	440	544	5.5	M5 screw into the steel
57	87	554	3.4	gutter-R
58	614	554	3.4	gutter-L
59	87	614	3.4	gutter-R
60	614	614	3.4	gutter-L
61	134	619	9.6	3/8 dowel
62	566	619	9.6	3/8 dowel
63	69	634	5.5	M5 insert screw
64	631	634	5.5	M5 insert screw
65	260	664	5.5	M5 screw into the steel
66	440	664	5.5	M5 screw into the steel
67	87	674	3.4	gutter-R
68	614	674	3.4	gutter-L
69	327	705	6.5	M6 screw into a pad
70	373	705	6.5	M6 screw into a pad
71	311	712	5.5	M5 screw into the steel
72	389	712	5.5	M5 screw into the steel
73	327	718	6.5	M6 screw into a pad
74	373	718	6.5	M6 screw into a pad
75	87	734	3.4	gutter-R
76	614	734	3.4	gutter-L
77	21	744	4.5	cleat

#	x	y	Hole	For
78	213	743	5.5	M5 insert screw
79	487	743	5.5	M5 insert screw
80	680	744	4.5	cleat
81	327	748	6.5	M6 screw into a pad
82	373	748	6.5	M6 screw into a pad
83	64	772	3.4	corner-R
84	637	772	3.4	corner-L
85	200	774	5.5	M5 insert screw
86	500	774	5.5	M5 insert screw
87	80	822	4.5	cleat
88	260	822	4.5	cleat
89	440	822	4.5	cleat
90	620	822	4.5	cleat

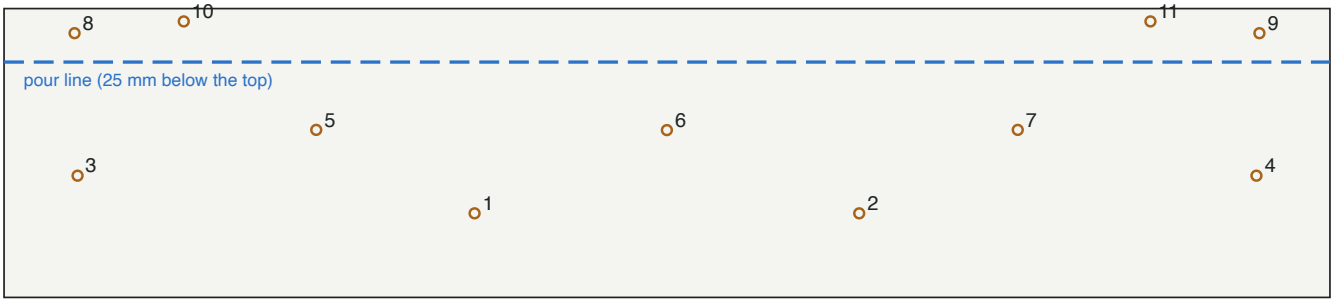
Base front wall



0,0 · 620 x 135 mm · seen from the cast side

#	x	y	Hole	For
1	220	39	4.5	draw-pocket-L
2	400	39	4.5	draw-pocket-R
3	34	57	4.5	corner-L
4	586	57	4.5	corner-R
5	146	78	4.5	bevel-end
6	310	78	4.5	bevel-end
7	474	78	4.5	bevel-end
8	33	124	4.5	corner bracket
9	588	124	4.5	corner bracket
10	84	129	4.5	leg holder
11	536	129	4.5	leg holder

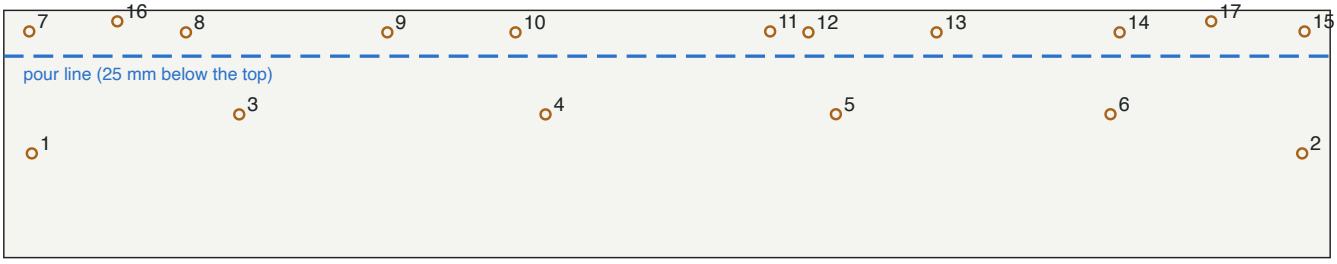
Base rear wall



0,0 · 620 x 135 mm · seen from the cast side

#	x	y	Hole	For
1	220	39	4.5	draw-pocket-R
2	400	39	4.5	draw-pocket-L
3	34	57	4.5	corner-R
4	586	57	4.5	corner-L
5	146	78	4.5	bevel-end
6	310	78	4.5	bevel-end
7	474	78	4.5	bevel-end
8	33	124	4.5	corner bracket
9	588	124	4.5	corner bracket
10	84	129	4.5	leg holder
11	536	129	4.5	leg holder

Base side wall, left

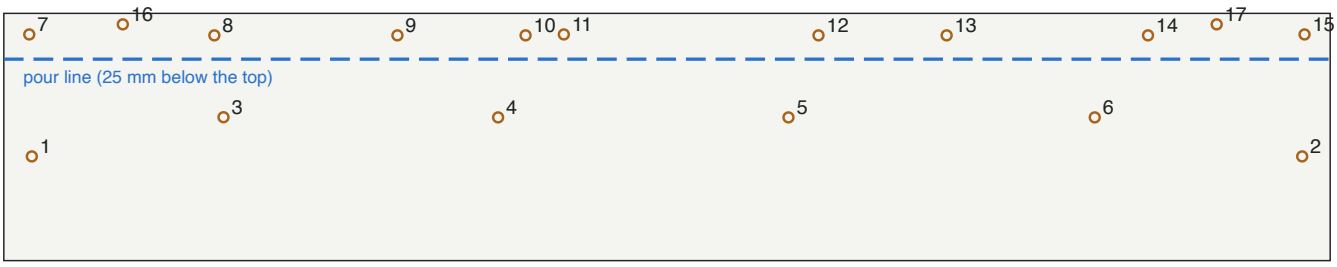


0,0 · 724 x 135 mm · seen from the cast side

#	x	y	Hole	For
1	15	57	4.5	corner-L
2	709	57	4.5	corner-L
3	129	78	4.5	bevel-L
4	296	78	4.5	bevel-L
5	454	78	4.5	bevel-L

#	x	y	Hole	For
6	604	78	4.5	bevel-L
7	14	124	4.5	corner bracket
8	100	123	4.5	saddle
9	210	123	4.5	saddle
10	280	123	4.5	saddle
11	419	124	4.5	corner bracket
12	440	123	4.5	saddle
13	510	123	4.5	saddle
14	610	123	4.5	saddle
15	711	124	4.5	corner bracket
16	62	129	4.5	leg holder
17	660	129	4.5	leg holder

Base side wall, right

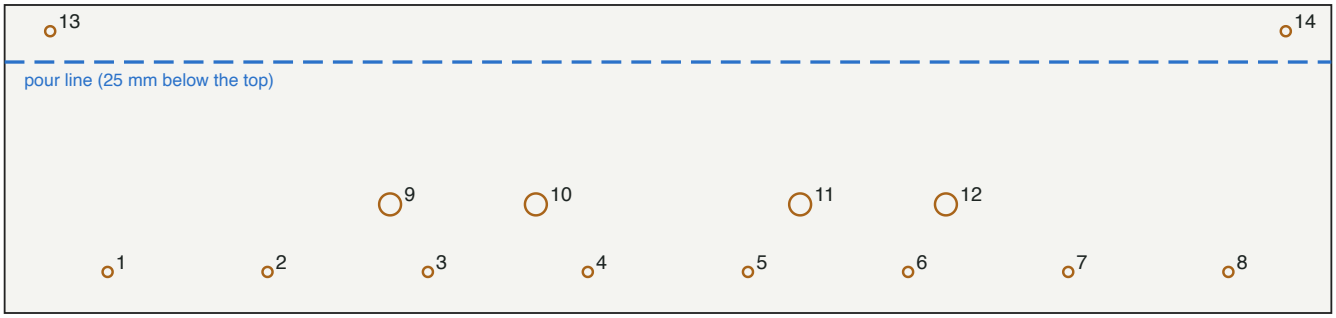


0,0 · 724 x 135 mm · seen from the cast side

#	x	y	Hole	For
1	15	57	4.5	corner-R
2	709	57	4.5	corner-R
3	120	78	4.5	bevel-R
4	270	78	4.5	bevel-R
5	429	78	4.5	bevel-R
6	596	78	4.5	bevel-R
7	14	124	4.5	corner bracket
8	115	123	4.5	saddle
9	215	123	4.5	saddle
10	285	123	4.5	saddle
11	306	124	4.5	corner bracket
12	445	123	4.5	saddle
13	515	123	4.5	saddle

#	x	y	Hole	For
14	625	123	4.5	saddle
15	711	124	4.5	corner bracket
16	65	129	4.5	leg holder
17	662	129	4.5	leg holder

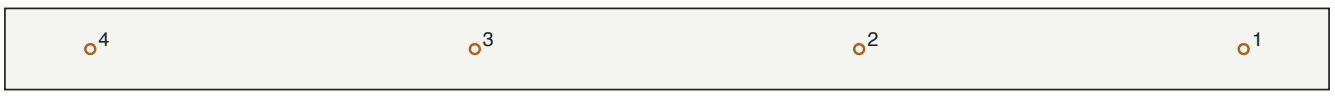
Base partition



0,0 · 582 x 135 mm · seen from the cast side

#	x	y	Hole	For
1	45	18	3.2	split-gasket-groove
2	115	18	3.2	split-gasket-groove
3	186	18	3.2	split-gasket-groove
4	256	18	3.2	split-gasket-groove
5	326	18	3.2	split-gasket-groove
6	397	18	3.2	split-gasket-groove
7	467	18	3.2	split-gasket-groove
8	537	18	3.2	split-gasket-groove
9	169	48	9.6	3/8 dowel pin (split)
10	233	48	9.6	3/8 dowel pin (split)
11	349	48	9.6	3/8 dowel pin (split)
12	413	48	9.6	3/8 dowel pin (split)
13	20	124	4.5	corner bracket
14	562	124	4.5	corner bracket

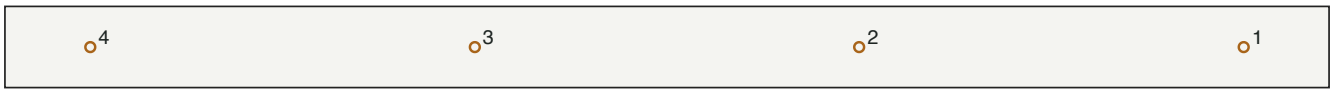
Base cleat, front



0,0 · 620 x 38 mm · seen from the outside (top)

#	x	y	Hole	For
1	580	19	4.5	cleat
2	400	19	4.5	cleat
3	220	19	4.5	cleat
4	40	19	4.5	cleat

Base cleat, rear



0,0 · 620 x 38 mm · seen from the outside (top)

#	x	y	Hole	For
1	580	19	4.5	cleat
2	400	19	4.5	cleat
3	220	19	4.5	cleat
4	40	19	4.5	cleat

Base cleat, left



0,0 · 724 x 38 mm · seen from the outside (top)

#	x	y	Hole	For
1	684	19	4.5	cleat
2	470	19	4.5	cleat
3	255	19	4.5	cleat
4	40	19	4.5	cleat

Base cleat, right

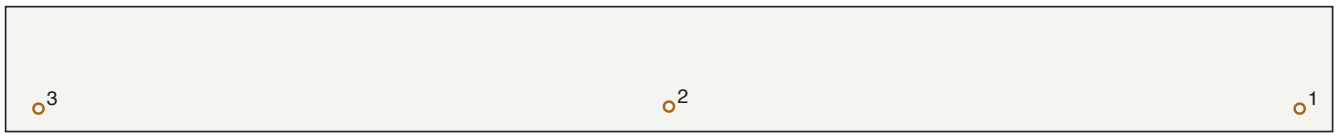


0,0 · 724 x 38 mm · seen from the outside (top)

#	x	y	Hole	For
1	684	19	4.5	cleat
2	470	19	4.5	cleat
3	255	19	4.5	cleat

#	x	y	Hole	For
4	40	19	4.5	cleat

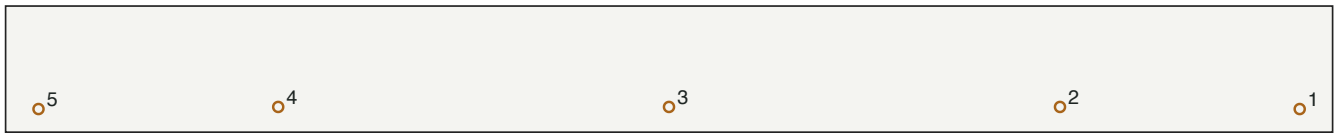
Base bridge y=-230



0,0 · 633 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	618	11	4.5	saddle
2	317	12	4.5	core post
3	16	11	4.5	saddle

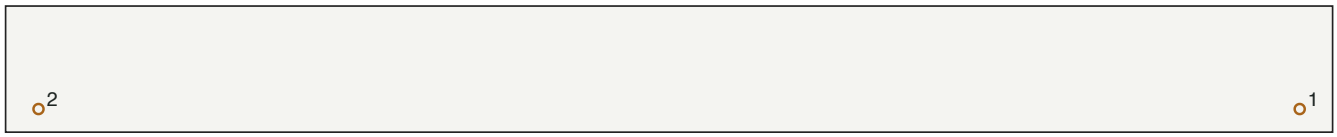
Base bridge y=-130



0,0 · 633 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	618	11	4.5	saddle
2	503	12	4.5	core post
3	317	12	4.5	core post
4	130	12	4.5	core post
5	16	11	4.5	saddle

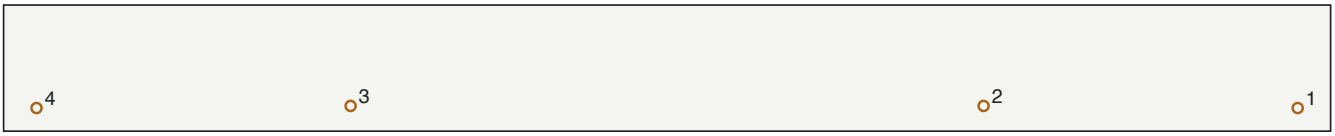
Base bridge y=-60



0,0 · 633 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	618	11	4.5	saddle
2	16	11	4.5	saddle

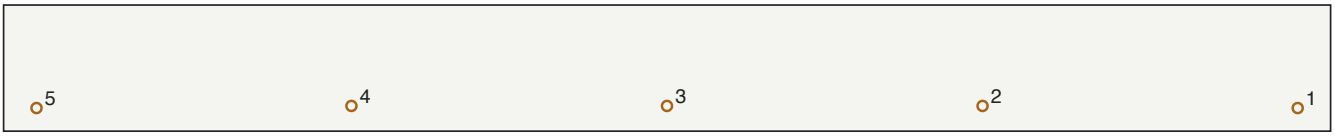
Base bridge y=100



0,0 · 633 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	618	11	4.5	saddle
2	468	12	4.5	core post
3	166	12	4.5	core post
4	16	11	4.5	saddle

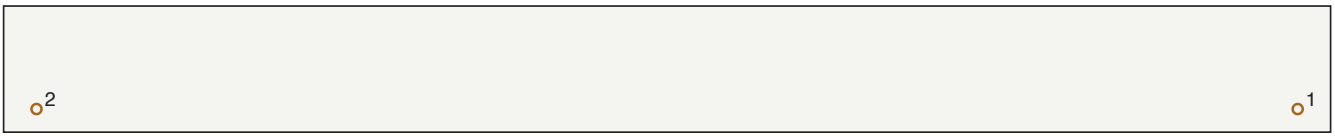
Base bridge y=170



0,0 · 633 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	618	11	4.5	saddle
2	467	12	4.5	core post
3	317	12	4.5	core post
4	166	12	4.5	core post
5	16	11	4.5	saddle

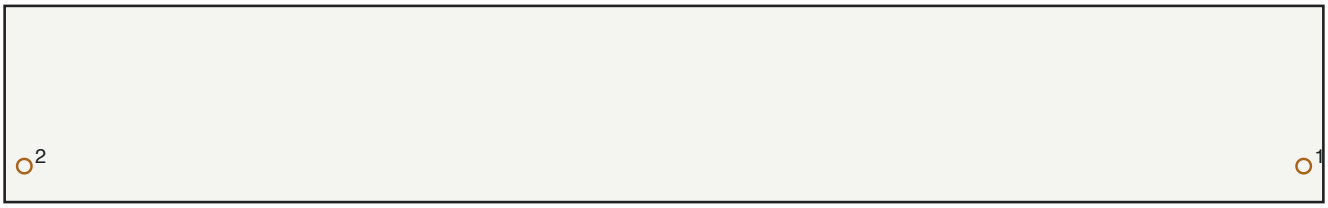
Base bridge y=280



0,0 · 633 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	618	11	4.5	saddle
2	16	11	4.5	saddle

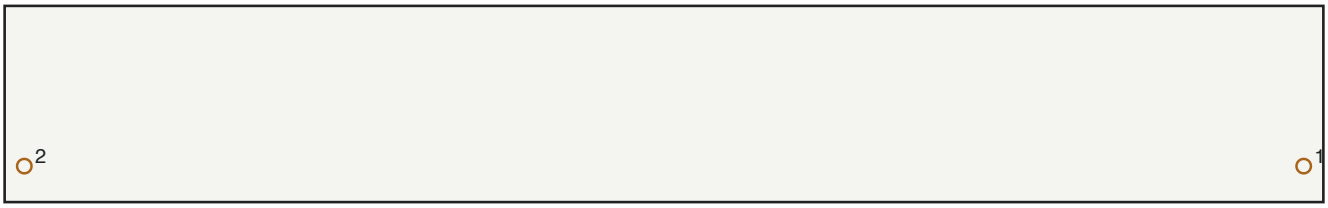
Base leg bridge y=-280



0,0 · 402 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	396	11	4.5	leg holder
2	6	11	4.5	leg holder

Base leg bridge y=318



0,0 · 402 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	396	11	4.5	leg holder
2	6	11	4.5	leg holder